



AER1416: Numerical Methods for Uncertainty Quantification

University of Toronto Institute for Aerospace Studies
<http://utias.utoronto.ca/~pbn>

Assignment 3 (25pts)

Due April 29, 2026

1. **(8pts)** Formulate a *non-intrusive* polynomial chaos projection scheme for approximating the first two statistical moments of the solution of problem **P3**. Implement your scheme in MATLAB/Python for a first-order and second-order PC-Legendre basis¹ and compare your results to those obtained in previous assignments.
2. **(7pts)** Formulate a polynomial chaos based *intrusive* stochastic Galerkin projection scheme for approximating the first two statistical moments of the solution of problem **P2**.²
 - **(5pts)** Present a detailed mathematical derivation of the steps involved in the stochastic Galerkin projection scheme.
 - **(2pts)** Formulate a numerical scheme to efficiently solve the equations resulting from the stochastic Galerkin projection scheme. You are not expected to implement your numerical scheme.
3. **(10pts)** Study the paper by Hennig, Osborne, and Girolami, “Probabilistic numerics and uncertainty in computations,” *Proc. R. Soc. A* 47120150142 <http://doi.org/10.1098/rspa.2015.0142>. Your response should be 3-4 pages in length (not smaller than 10pt font with 1-inch margins).
 - (a) The authors show how classical numerical methods can be reinterpreted as statistical inference procedures. Mathematically discuss this viewpoint within the context of uncertainty quantification.
 - (b) Choose one of the numerical tasks discussed in the paper (integration, linear algebra, optimization, or differential equations) and analyze how the probabilistic numerics approach could transform our understanding of this task. How might this perspective influence algorithm design in your chosen area?

¹For a uniformly distributed random variable, $\theta \sim \mathcal{U}(-1, 1)$, the Legendre polynomial chaos basis (up to second-order) is given by $\{1, \theta, \frac{1}{2}(3\theta^2 - 1)\}$. You can take a tensor product of these univariate basis functions to obtain a multivariate basis.

²The examples in these papers may be useful while formulating your numerical scheme: (1) Xiu and Karniadakis, “The Wiener-Askey polynomial chaos for stochastic differential equations,” *SIAM J. Sci. Comput.*, 24(2), 619-644, 2002. https://www.sci.utah.edu/~dxiu/Papers/XiuK_SISC02.pdf, and (2) Lucor and Karniadakis, “Adaptive Generalized Polynomial Chaos for Nonlinear Random Oscillators,” *SIAM Journal on Scientific Computing*, 26(2), 720-735, 2004 <https://doi.org/10.1137/S1064827503427984>.

- (c) Consider how the probabilistic numerics framework could complement or enhance one of the following methods we studied: control variates, quasi-Monte Carlo, importance sampling, sparse quadrature or polynomial chaos expansions. What theoretical or practical advantages might emerge from such integration?

Submission guidelines

- Upload an **electronic copy** of your report in **PDF format** along with your code and input files (if any) to Quercus.